graphics/Logo-schwarz.pdf

# Design Games

## [Draft: No ‘naustrian’ text for polylingual ‘Subtitle’.]

BACHELORARBEIT

zur Erlangung des akademischen Grades

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eingereicht von

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Betreuung: Pretitle Hilda Tellioğlu, Posttitle

Wien, 30. September 2026

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|---|---|

graphics/Logo-schwarz.pdf

# Design Games

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BACHELOR’S THESIS

submitted in partial fulfillment of the requirements for the degree of

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in

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by

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to the Faculty of Informatics

at the TU Wien

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Vienna, September 30, 2026

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Sami Pasquali

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Wien, 30. September 2026

Sami Pasquali

# Danksagung

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# Acknowledgements

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# Kurzfassung

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# Abstract

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CHAPTER 1

# Design Games

[Braa] describes Design games as a way of organizing participation in design projects. [TPT] notes that they (Design Games) provide a starting point for facilitating user participation and contribution.

## 1.1 History

## 1.2 Connection to Games

## 1.3 Categories and Classification of Design Games

## 1.4 Mechanics of Design Games

Throughout the literature review a number of reoccuring mechanics have been identified. The mechanics are subsequently clustered in 4 Clusters of effects they have on co-design sessions using design games. Many mechanics can be attributed to multiple effects and are therefore included in more than one cluster. Mechanics are ordered alphabetically.

### 1.4.1 Break existing habits

Breaking existing habits is a shift from everday logical thinking to generative, imaginative and lateral thinking. It grounds abstract thinking into a concrete reality so participants aren't overwhelmed by a blank canvas and overcomes potential power relations participants are part of usually.

#### Chance

Element of chance stems from the surrealist movement. "Games are often based on a wish to get out of habits, experience new and be taken by surprise" [Brab]. This might

come in the form of a dice, used to decide which topic or question to work on next.

### Combining elements that do not fit together

People can be triggered out of their normal thinking patterns by, sometimes unusual, associations, suggestions or actions writen on a card [RW]

### Element of fun & Ludic drive

The essence of play lies in the intensity and absorption of players. It is referred to as the "Fun element". This absorption has to be based on voluntary participation and acceptance of the game rules [BMB].

### Facilitation

A facilitator can leave out elements to increase the focus more on a specific topic or influence which topic or question is to be worked on next [KKM]

### Game rules

When playing, participants impersonate new roles that are defined by rules. Rules define what goal players strive to achieve, how success is defined and what actions are allowed. Thus, rules establish creative freedom by excluding ordinary life, while simultaneously forming its boundaries [FGKM].

A defined winning condition and rules to guide argumentation are found to make it easier for participants to reach consensus, which is described as more troublesome in classic co-design workshops with strangers [ZPZ$^+$].

### Involving stakeholders

Participants should come from different backgrounds and explore cooperation through playing. Design Games establish out-of-the-ordinary communication that breaks existing routines and communication habits, allowing new problem solving approaches to emerge [FGKM]

### Magic circle

The magic circle is a temporary sphere of activity in which the laws of ordinary life do not count. It invites participants to think beyond the ordinary and envision alternative solutions [VM]. [FGKM]

### Presupplied input

How topics are framed by the supplied game material matters to the outcome of the game [BMB].

#### Tangible game material

Through Concept Design Games it has been found that Game Pieces and Game Boards can be abstracted to eliminate functional knowledge and former experiences participants bring with them [Braa]

### 1.4.2 Come up with creative ideas & Overcoming cognitive intertia

Autors talk about "scaffolding" to help non-designers access their creativity. Without some structure, people default to what they already know, or experience analysis paralysis.

#### Abstracted and ambiguous tasks

Help participants to propose new interpretations and alternative solutions to come up with creative ideas [VM], [BHK].

#### Boundary objects

Boundary objects make arguments tangible and help participants to focus [BM].

#### Chance

Elements of chance, for example through dice throwing, decide for the player which questions are to be worked on next [HM]. Another method is randomly drawing cards, which then provide a starting point for discussion or building a scenario [BHK]. Games often remain open-ended until the very end, allowing for new twists through elements of chance and luck [FGKM].

#### Combining elements that do not fit together

Steming from the surrealist movement, some creativity techniques are based on combining elements which normally do not fit together [Braa]. Cards allow to easily violate cateogries of information by physically placing them different, which allows the brains creativity to come in and repair them in an ad-hoc manner [RW].

#### Design Context

Design Games are specifically designed for a specific context in mind. Depending on this context, different game qualitites are desirable, and different participants are needed [VM], [GCAG], [FGKM].

#### Establishing an informal atmosphere

Design Games are usually played in an informal atmosphere, which is shown to be the most productive in creative work [Braa].

### Facilitation

A facilitator can lead the workshop and present the agenda, game pieces used and rules to follow. Facilitators can also take an active role and drive the process forward by making suggestions, as well as encouraging players to make moves [BM]. Further, dividing the design session into sequential phases has shown to drive the process forward in a more natural way, which was found similar to divergent and convergent design thinking [BHK].

### Game rules

Rules set the boundaries for whats possible and also structure the play of the game. The shift from the goals of the design process to the rules of a game makes it easier to generate design moves. Further, restrictions on idea generation is shown to improve creative outcome [BM].

### Open-endedness

Open-ended and ambiguous tasks help participants to propose new interpretations and alternative solutions [VM]. Game material, e.g. cards, can supply dedicaded space where participants can add their own notes [HM], [NETM$^+$]. Lastly, design games themselves are open-ended and open for evolution [FGKM]

### Presupplied input

A question, e.g. from cards, might be used to spark further discussion [Braa], [NETM$^+$]. Predesigned material can also help in setting the frame and provide starting point for the design collaboration [VM], [GCAG]. Graphically represented themes on, e.g. cards, can guide conversations throughout the session [HM]. Cards can further be used to summarize good design practises or domain knowledge and hint participants towards it during the design session [RW], [NETM$^+$]

### Scenarios

Design Games construct scenarios that are both concrete and flexible. They represent a particular interpretation of a design situation. [BM]. Scenario construction can be used for creating and testing design ideas with future users. Generic game pieces can be provided for that or users encouraged to produce their own [Braa].

### Split participants in groups

Participants may play the same game in parallel, split in groups. Thus each group comes to their own conclusion [EBMV]. Another approach is to let participants first individually create ideas to then share them with the rest of their team or between groups [BHK].

#### Tangible game material

Visual and tangible components in design games are generative tools. They aim to create a platform for shared focus of attention and also provide documentation and reminder during and after the co-design session [VM]. Ease of moving around cards on the table and pairing them new enabled creation of links across themes and gain new perspectives into underlying causes of some challenges [HM].

#### Time keeping

Keeping track of time can improve structure of discussion and prompt the team to conclude their discussions [EBMV]. Time keeping also aids in equally distributing time between the different topics to discuss [EBMV]. In design games players often do not compete against each other, but might collaborate to "race against the clock". [FGKM].

#### Win condition

There is rarely competition against each other in design game. However, tension is a quality of design games and aids the creative process [VM]. Misalignment between goals - for example an individual one for the player and an overall one for the game - can also impede the success of the design game [KGPC], [FGKM].

### 1.4.3 Create common understanding

### 1.4.4 Learn about concepts participants hold & Build empathy

### 1.4.5 Overcome power relations & Lower communication barriers in a group setting

### 1.4.6 Discussion

Analysieren warum bestimmte Mechaniken in fast jedem Paper auftauchen.

## 1.5 Benefits

## 1.6 Problems and Issues

CHAPTER 2

# Examples of Design Game and their use in different domains

CHAPTER 3

# Design Games in Design Thinking

## 3.1 Opportunities for Design Game use

# Overview of Generative AI Tools Used

No AI tools were used for creating this paper.

# Übersicht verwendeter Hilfsmittel

Keine

# List of Figures

# List of Tables

# List of Algorithms

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